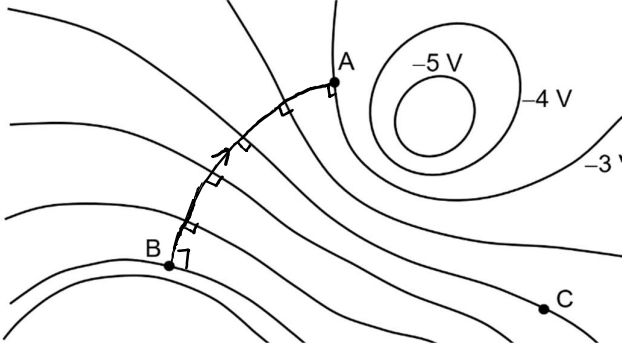
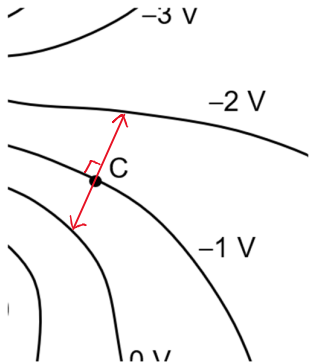


Qns	Answer	Marks
7a	Electric potential at a point is <u>the work done</u> (by an external agent) <u>per unit positive charge to move a small test charge from infinity to that point</u> .	B1
7bi	 The field line must be perpendicular to all the field lines from A to B. The direction (as indicated by the arrow) is from B to A. The field line must be reasonably smooth. –1 for each mistake (to a minimum of 0).	
7bii	By Principle of Conservation of Energy, Gain in Kinetic energy = Loss in Electric Potential Energy $\frac{1}{2} mv^2 - \frac{1}{2} mu^2 = e (V_i - V_f)$ $\frac{1}{2} (9.11 \times 10^{-31})v^2 - \frac{1}{2} (9.11 \times 10^{-31})(2.6 \times 10^6)^2 = -1.6 \times 10^{-19}(-3-2)$ $v = 2.9 \times 10^6 \text{ m s}^{-1} \text{ (} 2.92 \times 10^6 \text{)}$	C1 Statement C1 decrease in EPE A1
7biii	 Considering the distance between the nearest two equipotential lines around point C (0V and –2V) $d = 1.8 \text{ cm (accept 1.7 to 1.9cm)}$ $E = \Delta V/d = (2-0)/(1.8 \times 10^{-2})$ $F = eE = 1.6 \times 10^{-19} \times (2-0)/(1.8 \times 10^{-2})$ $= 1.8 \times 10^{-17} \text{ N}$	C1

		A1
Qns	Answer	Marks
8(a)(i)	In a polarised wave, the oscillations are along one direction only.	B1